



"X" stands for Excellence , Experimentation , Collaboration and The unknown

Hackathon Campaign Timeline

Launch – Oct 30 2025

- Hackathon Overview to Professors
- IDEA Cards
- Sharing forms to Professors

Form submission
phase – Oct 28– Nov
07 2025

- Colleges to form teams
- Choose ID cards
- Submit forms

Kick start the
Hackathon – Nov
11 2025

- Meet the teams
- Hackathon overview
- ID card elaboration
- Introduce the mentors

Innovation Phase –
Nov 12 to Feb 11 2026

- Weekly Connect
with mentors
- Update on progress

Evaluation
Phase – Feb
12 –
Feb 20 2026

Finale
Feb 23, 2026

- Top 10 teams to
pitch in front of SE
Leadership
- Selection of winners

Accelerating Industrial Automation solutions with Low-Code/No-Code Platforms

Mentor: Sudheer Madireddy

Idea Card ID: 01

Problem Statement

- Despite rapid advancements in industrial automation, many manufacturing and operations teams remain constrained by legacy systems, manual workflows, and fragmented data sources. These limitations hinder agility, slow down innovation, and increase operational costs.
- Traditional automation approaches demand specialized technical skills and lengthy development cycles, making it challenging for frontline teams to respond swiftly to evolving business and production requirements.
- There is a critical need for accessible, flexible, and cost-effective low-code / no-code solutions that empower non-technical users to digitize processes, integrate disparate systems, and drive continuous improvement on the shop floor.

Prerequisites

- Accounts on platforms like MS Team (used for coordination).
- Access for SharePoint for updates and submissions
- Basic programming knowledge (languages like Python, JavaScript, etc.)
- Familiarity with development tools (IDEs, Git, APIs)
- Understanding of the theme (e.g., AI/ML, Industrial Automation, SDL, etc.)

Expected Outcome

The initiative / idea aims to showcase how low-code/no-code platforms can democratize industrial automation by enabling rapid development and deployment of solutions—without requiring deep programming expertise. Expected outcomes include:

- Seamless digitization of manual and paper-based workflows, improving efficiency and traceability.
- Real-time, customizable dashboards for enhanced visibility into machine performance, sensor data, and key production metrics.
- Automated alerts and notifications to proactively address equipment issues and minimize downtime.
- Easy integration with existing industrial systems (PLCs, SCADA, IoT devices) through intuitive connectors or APIs.
- Implementation of predictive maintenance and anomaly detection using visual, user-friendly tools.
- Solutions that are secure, scalable, and compliant with industrial standards—empowering plant managers, engineers, and technicians to innovate and adapt quickly to operational changes.

Target Audience

- Computer Science, AI/ML
- Electronics and Communication
- Instrumentation and Controls

Extract symbols from P&ID using generative AI / Agentic AI

Mentor: Padmaja Bodanapu
Idea Card ID: 02

Problem Statement

Digitizing information from P&ID documents is a time-consuming and resource-intensive process in HMI engineering. While some existing solutions leverage deep learning to automate this task, they often require significant customization and frequent retraining to remain effective. The proposed solution is designed to address these limitations and streamline the conversion process.

Prerequisites

- User should be able to process P&IDs (images, PDFs, scanned documents) using Multimodal LLM.
- Symbol and layout recognition using vision-language models.
- Good to have a self-learning feature of user given symbols
- Agentic workflow for iterative refinement and feedback incorporation
- The application should complete the process in a reasonable time frame.
- The symbol detection should be accurate, minimizing false positives and negatives.
- Application should be able to identify various types of symbols present in P&ID like valves, tags, vessels, motors and other equipment's etc.

Expected Outcome

In a typical project comprising more than 500 P&ID documents, users are required to identify various symbols present in those P&IDs. Develop an intelligent GenAI-powered agent that autonomously processes P&ID documents, understands symbols and layouts, and converts them into structured digital formats (like XML, XLS etc.) with minimal human intervention. Good to have a self-learning feature, where will give one symbols, and system should be able to identify similar symbols from P&ID. By automating this process, the solution enhances symbol identification accuracy, boosts productivity, and supports more informed decision-making in engineering and analysis workflows.

By leveraging multimodal GenAI and agentic workflows, the system can extract the symbols present in P&ID and learn from a variety of document formats and user feedback, reducing the need for user manual review and improving accuracy over time.

Target Audience

Computer Science, AI/ML
Electronics and Communication
Instrumentation and Controls
Chemical

Extract logic information from Control Narrative using generative AI / Agentic AI

Mentor: Padmaja Bodanapu
Idea Card ID: 03

Problem Statement

Control narratives are documents that outline the requirements and detailed descriptions of complex process control loops. However, extracting control logic from these narratives is often time-consuming and challenging, as engineers must frequently cross-reference multiple sections. Designing control logic for such complex loops can consume approximately 40% of the total software engineering effort.

Implementing Generative AI and Agentic AI to automate the extraction and design of control logic from control narratives will significantly reduce manual engineering effort, improve accuracy, and accelerate software development cycles—potentially cutting control logic design time by up to 40%

Prerequisites

- Ability to extract control logic of complex loops for uploaded Control Narrative.
- GenAI model to understand process different types are control narrative document effectively.
- AI agents to identify gaps in extracted control logic and assist users to regenerate logic and related information
- Contextual understanding to provide accurate and relevant responses based on the specific content
- An application allows user to be able to process the control narrative document in PDF form. It should give the required control logic related information in a reasonable time.

Expected Outcome

Automate the extraction of control logic from complex control narratives using Generative AI. Agentic AI agents validate, refine, and interact with engineers to ensure accurate and complete logic design. Identify key elements like control loop types, setpoints, interlocks, alarms, and sequences. Automatically generate control logic in structured formats (e.g., ladder logic, function block diagrams, or pseudo-code). Also, AI Agents can identify inconsistencies, missing logic, or ambiguous requirements and resolve them.

Application should be able to extract the relevant information from control narrative based on user given complex loop.

Ex. Control Logic Identification: Correctly identifies control elements (e.g., setpoints, logic gates and loop initialization).

Loop Mapping: Accurately maps relationships between control elements (e.g., feedback loops, cascaded loops).

Target Audience

Computer Science, AI/ML
Electronics and Communication
Instrumentation and Controls
Chemical

Smart Sensing of Metal ion concentration

Mentor: Satish Kumar Lekkala

Idea Card ID: 04

Problem Statement

In the metals and mining industry, precise real-time control of soluble metal ion concentrations in process streams—such as leach solutions, solvent extraction, and electrowinning—is critical for maximizing metal recovery. Traditional methods rely heavily on manual sampling and physical titration, which are time-consuming, manpower-intensive, and prone to delays in decision-making.

Development of a **real-time metal ion concentration analysis system** for both acidic and basic solutions. By leveraging advanced sensors, machine learning models, and integrated process control, the solution will eliminate the need for physical titration, enabling faster, more efficient, and adaptive control of metallurgical processes. This will significantly enhance operational efficiency, energy conservation and mineral processing operations.

Prerequisites

The sensors and materials used shall be chemically resistant to:

1. Strong acids and bases
2. Solvents
3. Corrosive chemicals

Results accuracy should be within minimum error tolerance.

- The system shall include automated calibration routines. Maintenance requirements shall be minimal and clearly defined

Expected Outcome

Develop or implement alternative economical analytical methods such as:

- Colorimetric analysis , Light refraction or reflection techniques , Optical sensors , Solution conductivity-based measurements , X-ray / Laser technologies etc if any....
- Consider any metal composition in acidic and basic environment.

Real-Time metal concentration detection

1. The system shall continuously monitor the concentration of metal ions in the solution.
2. The detection method shall provide results with minimal latency (preferably under 1 minute).

Target Audience

Metallurgy & Chemical Engineers, AI/ML
Electronics and Communication
Instrumentation and Controls

Smart Metal Ore Grinding and Hydrocyclone Control for Improved Recovery and Water Conservation

Mentor: Satish Kumar Lekkala

Idea Card ID: 05

Problem Statement

In mineral processing operations, particularly in grinding and classification circuits, inefficient control of ball mill grinding and hydrocyclone performance can lead to suboptimal metal recovery. Variability in hydrocyclone feed flows affects classification stability, resulting in inconsistent particle size distribution. Excessive water usage in grinding operations can dilute slurry density, impacting grinding efficiency and downstream separation processes. Additionally, non-uniform or overly coarse/fine particles contribute to flotation losses, reduced recovery rates, and increased energy consumption due to overgrinding.

These challenges collectively hinder process stability, increase operational costs, and reduce overall metallurgical performance in concentrator plants.

Prerequisites

Real-Time metal PSD Detection

1. The system shall continuously monitor the PSD from the ball mill outlet, hydrocyclone feed , overflow and under flow
2. The detection method shall be continuous and recovery of fine particles more in the hydrocyclone overflow with out slipping to underflow

Expected Outcome

Real-time particle size distribution (PSD) monitoring in ball mills and hydrocyclones through advanced sensor technologies and machine learning models, eliminating manual sampling and reducing overgrinding. Solutions should enable continuous hydrocyclone performance tracking to prevent fines from recirculating to the ball mill, ensuring stable classification and improved recovery.

It is expected that by achieving the target particle size faster and avoiding prolonged milling, overall power consumption will decrease significantly, while water losses due to excessive dilution will be minimized. This will lead to improved energy efficiency, reduced water usage, and a more sustainable and cost-effective grinding operation.

Target Audience

Metallurgy & Chemical Engineers
Computer science, AI/ML
Electronics and Communication
Instrumentation and Controls

AI powered code review

Mentor: SaiAshokKumar Dogga
Idea Card ID: 06

Problem Statement

The developer in VSCode / Visual Studio spends a good amount of time developing the coded logic and even uses GitHub Copilot for development assistance. As part of Organizational mandate, the developer now has to get the development code peer reviewed. This involves a workflow with tools such as SmartBear and an Org mandated checklist which involves waiting time with the peer. AI should ease this pain workflows and waiting time.

Prerequisites

- There should be a standards checklist-based input to Code review
- Code reviews should provide recommendations based on checklist
- Code reviews should have facility to auto implement
- Code reviews should be governed by checklists
- Code reviews should have a tangible link to be used in ADO Tasks for audit purposes

Expected Outcome

The code reviews should be built-in to the VSCode or Visual Studio.
The code reviews should provide recommendations and corrective actions based on check lists

Target Audience

Computer science, AI/ML
Electronics and Communication

AI powered unit testing

Mentor: SaiAshokKumar Dogga
Idea Card ID: 07

Problem Statement

The developer in VSCode / Visual Studio spends a good amount of time developing the coded logic and even uses GitHub Copilot for development assistance. As part of Organizational mandate, the developer now has to get the code / functionality unit tested and will need to use a tool such as Junit. This again involves coding. AI should ease this pain of coding in unit tests.

Prerequisites

- There should be a prompt-based input to Unit testing
- Unit test reports should get generated providing clear Pass/Fail results
- The results (Pass/Fail) should be accompanied by a recommendation of improvement/fixes
- Coverage should be of the order of 1 unit test per 10–50 lines of production code.
- Each test should execute in < 100ms for fast feedback loops.

Expected Outcome

The Unit tests should be built-in to the VSCode or Visual Studio.
The Unit tests should generate reports
The Unit tests should run independent of user integration which means an ideal Unit test should generate its own assertion case data

Target Audience

Computer science, AI/ML
Electronics and Communication

AI based graphics along with process parameters

Mentor: SaiAshokKumar Dogga

Idea Card ID: 08

Problem Statement

Implementation of logics using customer data is leading to some uncertainties at sites, which causes fatal errors. To avoid it a smart logic builder is required to generate code using AI.

Prerequisites

- AI based smart logic builder should accept control logic written in typical control philosophy of SFC, FBD, Ladder, ST or IL
- The output of the smart logic builder should be graphic files that can be imported into standard HMI tools.
- The graphic files should be ready to use with control tags.

Expected Outcome

Implementing Logics(SFC, FBD, Ladder(PLB), ST IL) along with GUI for any process there will be a control logic, this control logic will be the input to AI, and AI will process data and create logic along with graphics in FoxDraw/CHMI along with process parameters. This will help in building logic without the efforts of field engineers, and reduce manual errors, fault occurrences will be minimal, tight closure of interlocks, and alarm systems

Target Audience

Computer science, AI/ML
Instrumentation and Control
Electronics and Communication

Safety Instrumented System Failure Predictor

Mentor: Karthikeyan Nagarajan

Idea Card ID: 09

Problem Statement

A Safety Instrumented System (SIS) has a probability of failure on demand (for SIL 3) between 10^{-3} and 10^{-4} in the field. The system rarely fails in the field. If it fails, the cost of safety associated with the failure is high. Currently there exists no reliable method of predicting the Safety System Failure.

Prerequisites

- Ability to extract events from the system events records.
- GenAI model to understand and process different types of events effectively.
- AI agents to identify key events and shall predict the SIS failure 72 hours ahead of the failure
- Contextual understanding to provide accurate and relevant responses based on the specific content

Expected Outcome

Automate the analysis of the system events recorder, identify a pattern of the events which could potentially cause the SIS failure at least 72 hours ahead of the failure using Gen AI. Agentic AI agents shall interact with engineers about the potential failure threat well ahead of time.

- Results accuracy should be within minimum error tolerance.
- The output shall be a prediction with valid system of events leading to the potential failure.

Target Audience

Electronics
Instrumentation and Control
Robotics
Electrical
Computer Science

Learn and build Safety Application program from existing files

Hariprasad Parthasarathy

Idea Card ID: 10

Problem Statement

Safety narratives are documents that outline the requirements and detailed descriptions of Safety logics. Safety logics are more or less standardized for a given application. Extracting the information from documents and PIDs will lead to errors and certification issues. However, extracting safety logic from existing certified programs for a given application will add a lot of value in terms of engineering time and cost.

Implementing Generative AI and Agentic AI to automate the creating of safety logics from the existing safety control program will significantly reduce manual engineering effort, improve accuracy, and accelerate safety lifecycle development and helps a lot with certification also.

Prerequisites

- Ability to extract safety logic from the existing Safety Control programs . This will include the configuration parameters also.
- GenAI model to understand and process different types of configuration and safety control programs effectively.
- AI agents to identify gaps in extracted control logic and assist users to regenerate logic and related information
- Contextual understanding to provide accurate and relevant responses based on the specific content

Expected Outcome

Automate the extraction of safety logic from existing TS1131 project files (IEC 61131-3 language – ST/FBD/ LD, HW configuration/tag definitions) in order to be able to autogenerate safety control programs per requirements for an application using Generative AI. Agentic AI agents validate, refine, and interact with engineers to ensure accurate and complete logic design and program. Identify key elements like safety function, setpoints, interlocks, alarms, shutdown logic and sequences. Automatically generate safety logic in structured formats (e.g., ladder logic, function block diagrams, Structured Text). Also, AI Agents can identify inconsistencies, missing logic, or ambiguous requirements and resolve them.

Application should be able to extract the relevant information from existing TS1131 application program

Ex. prompt to an AI “create a project file for system v12.2 with 2 chassis configured with 3 UIO modules, with two programs. The programs will do” The project files are primarily XML

Target Audience

Electronics
Instrumentation and Control
Robotics
Electrical
Computer Science

Implement C & E in Controller using ANN

Smruthi Ramesh
Idea Card ID: 11

Problem Statement

Safety narratives are documents that outline the requirements and detailed descriptions of Safety logics. This includes Cause and Effect diagram which is a matrix-based document that shows the relationship between process deviations (causes) and the resulting actions or equipment trips (effects). It serves as the basis for safety logics to be built in Safety Systems with input, output and timing requirements. Extracting the information from documents and PIDs will lead to errors and certification issues. Configuring the safety controller for the Cause and Effect will help with accuracy and cost benefits.

Prerequisites

- Ability to configure an ANN in a safety controller
- Implement the Input / Output and Time requirements of the Cause-and-Effect diagram.
- Contextual understanding to provide accurate and relevant responses based on the specific content

Expected Outcome

Configure the Safety controller using an ANN. This configuration will help to implement the input / output / timing requirements defined in a Cause and Effects Matrix. The format of the cause and effects matrix should be per IEC 62881

Target Audience

Electronics
Instrumentation and Control
Robotics
Electrical
Computer Science

Automatic Update to Technical Documents

Mentor: Sachin Patil

Idea Card ID: 12

Problem Statement

A Product release requires an update to the Technical documents (such as user documents, planning and installation guide, safety consideration guide). Maintaining these documents and updating the required documents to match the implemented design has become a tedious and time-consuming activity Release Over Release for every product.

Prerequisites

Technical Document update

1. The tool shall check the implemented GUI for every product, compare with the latest as available in technical documents and updates with standard documentation practices.
2. Agents to identify gaps in extracted GUI / content and assist users understand the difference and update them accordingly.
3. Role based privilege and access.
4. A log file that summarizes the changes made with relevant content.

Expected Outcome

To design and implement a platform agnostic tool that compares all the GUI images from the software application with the images in the user documentation; identifies which ones /where have changed; and updates all relevant documents and provides with the summary of changes for every document that has been updated.

- Results accuracy should be within minimum error tolerance.
- The developed tool should be modular and maintainable with plugin features and standard interface capability.

Target Audience

Electronics
Instrumentation and Control
Robotics
Electrical
Computer Science

Evaluation Criteria

Differentiation:

- Team presents a unique approach in solving
- Solution showcases creative thinking and innovation

Technology Readiness:

- Adoption of new technologies
- Feasibility to Implement the technology

Solution Readiness:

- Meeting the acceptance criteria in the ID card
- Demo able Minimum Viable Product/Solution

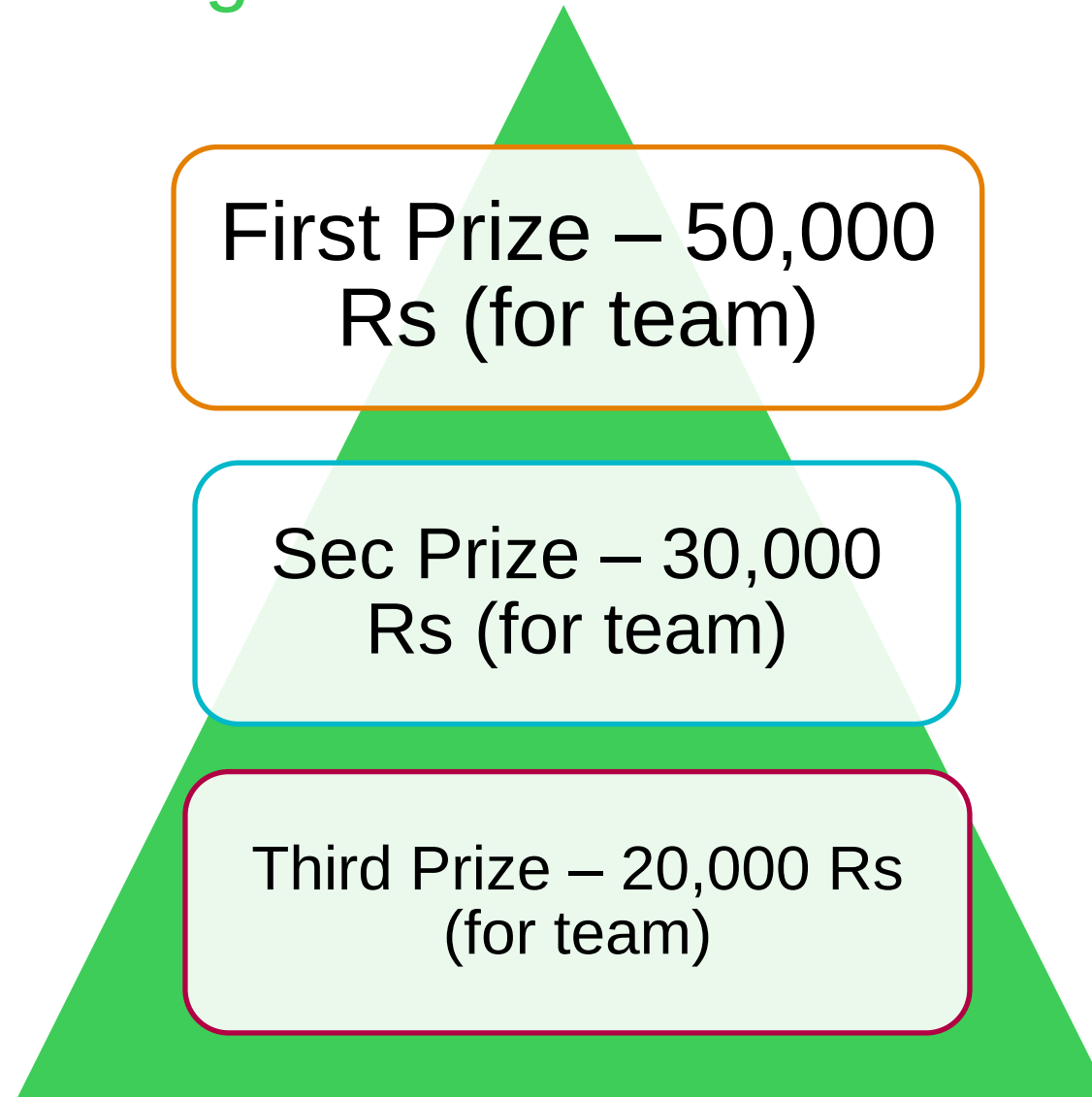
Scalability & Sustainability:

- Solution has potential for scalability and broader application



Call for action:
Bring your great
minds together! –
Form teams and
Register

Rewards and Recognitions



- Participation Certificate
- Winner Certificate
- One day Visit to SE office and Lab

FAQs

- University / College can nominate their teams to any number of ideas
- University / College can nominate two teams for a single idea provided they have nominations for atleast four other distinct ideas
- Weekly connect with mentors is mandatory (Updates and clarifications to be done only through this)
- Teams should keep posting the updates (code/PPT) in given sharepoint location
- Team size – 2 to max 5

Post further clarifications to :

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